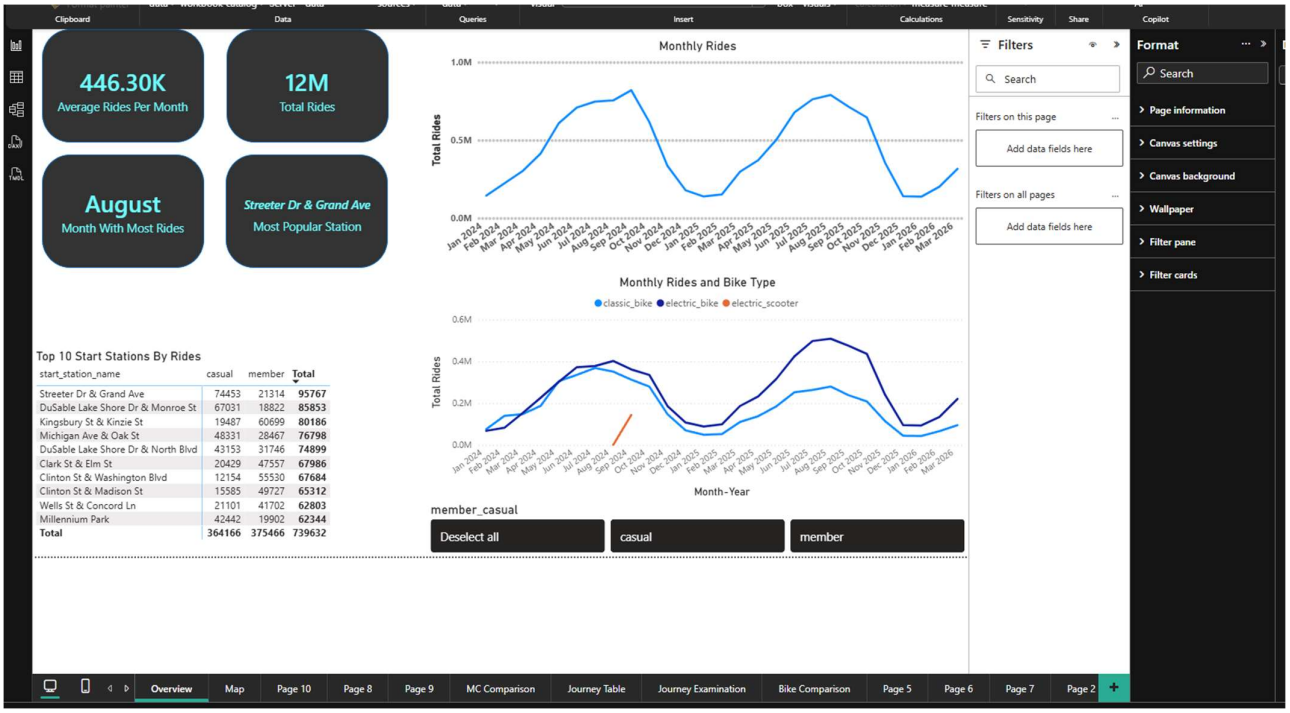
Cyclistic Analysis — Key Findings

A visual summary of the completed analysis, with screenshots used as evidence for the main findings. The focus is on what the data shows, how the rider groups differ, and the potential business implications.

1. Overall Seasonal Behaviour

The overview provides the broadest picture of the dataset before looking at the individual rider groups, stations, or journey patterns. Ride volume follows a clear seasonal cycle across the period analysed: activity rises through spring, reaches its highest levels during the warmer months, and then falls sharply through autumn and winter. August is the month with the highest total ride volume in the overview, while the winter months show substantially lower activity.

The same seasonal pattern is visible across the monthly ride and bike-type trends. Both member and casual activity rises and falls with the overall seasonal cycle, although the later analysis shows that the two groups differ in how and when they use the service. This overview therefore establishes the baseline for the more detailed comparisons that follow.



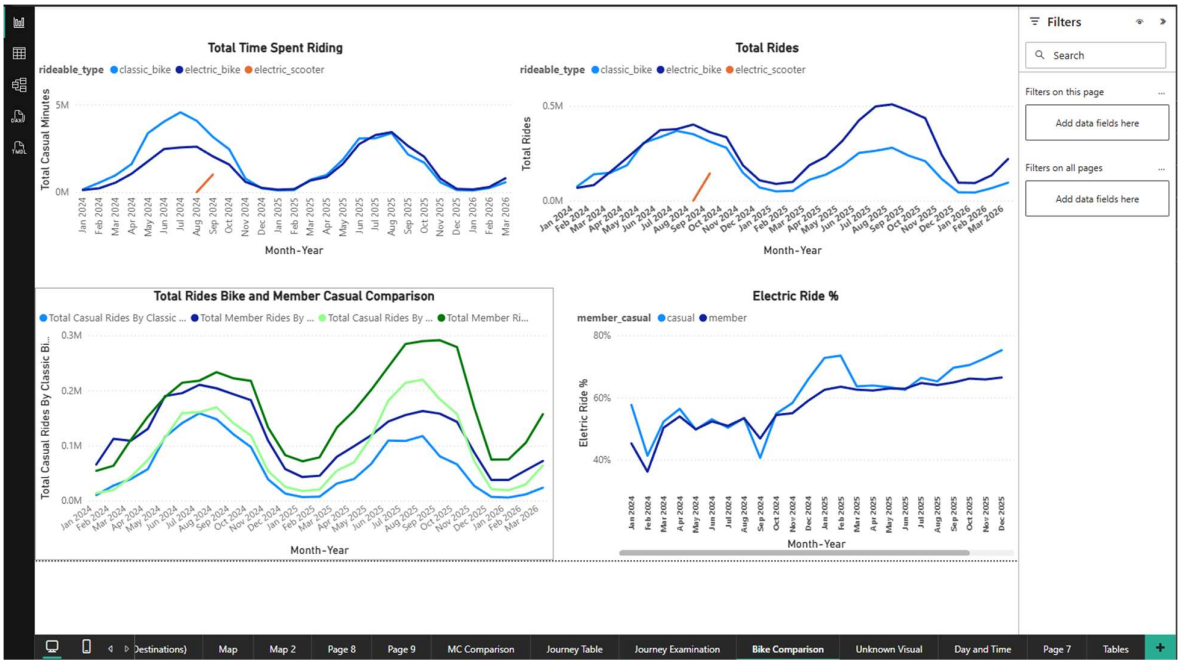
Power BI overview dashboard showing the overall seasonal pattern in ride volume, monthly ride activity by bike type, and the highest-volume start stations.

2. Rideable Type and Electric Bike Adoption

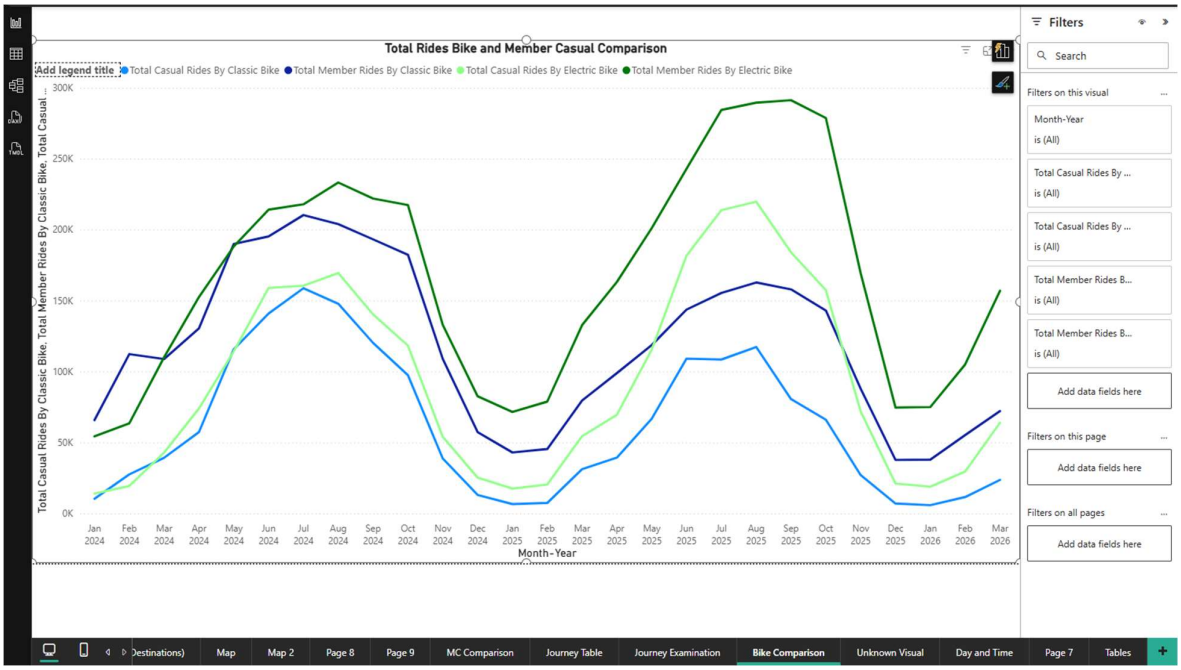
The earlier analysis identified a substantial increase in electric-bike usage during 2025 and into early 2026. The additional member-versus-casual comparison shows that the increase was not driven by only one rider group: both members and casual riders increased their use of electric bikes.

Casual riders generally have a slightly higher electric-bike share than members, particularly from early 2025 onward. The bike-type comparison also shows that electric-bike rides tend to account for more riding time than classic-bike rides. This makes electric-bike availability a potentially relevant

membership benefit to investigate, but the analysis does not contain pricing, revenue, or conversion data and therefore does not establish that restricting or discounting electric bikes would increase memberships.



Bike Comparison.png



Bike Comparison 2.png

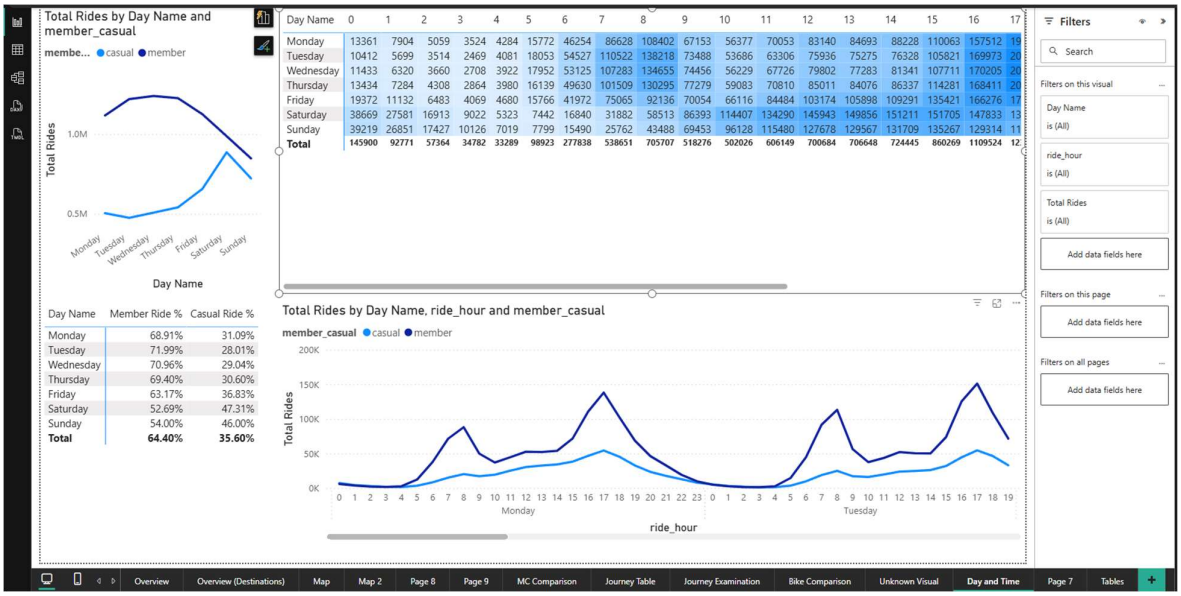
3. Member and Casual Behaviour by Day and Time

The day-of-week analysis shows a clear difference in the composition of rides. Members account for the majority of rides on every weekday, while the member-versus-casual gap becomes much smaller at weekends.

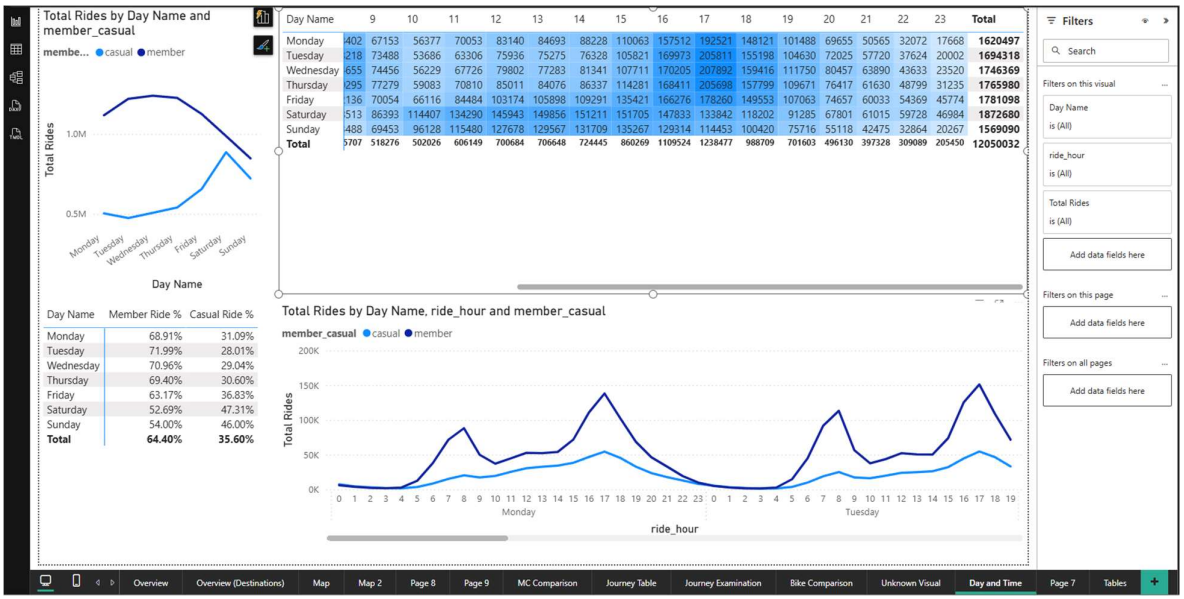
Across the full period, the daily composition was approximately 64.4% member and 35.6% casual. Saturday was approximately 52.7% member and 47.3% casual, while Sunday was approximately 54.0% member and 46.0% casual.

The hourly analysis provides a more detailed picture. Members show strong weekday morning and late-afternoon/early-evening peaks, particularly around the middle of the working week. Casual rides become much more prominent during weekend daytime hours. Several Saturday and Sunday afternoon periods approach an even member/casual split.

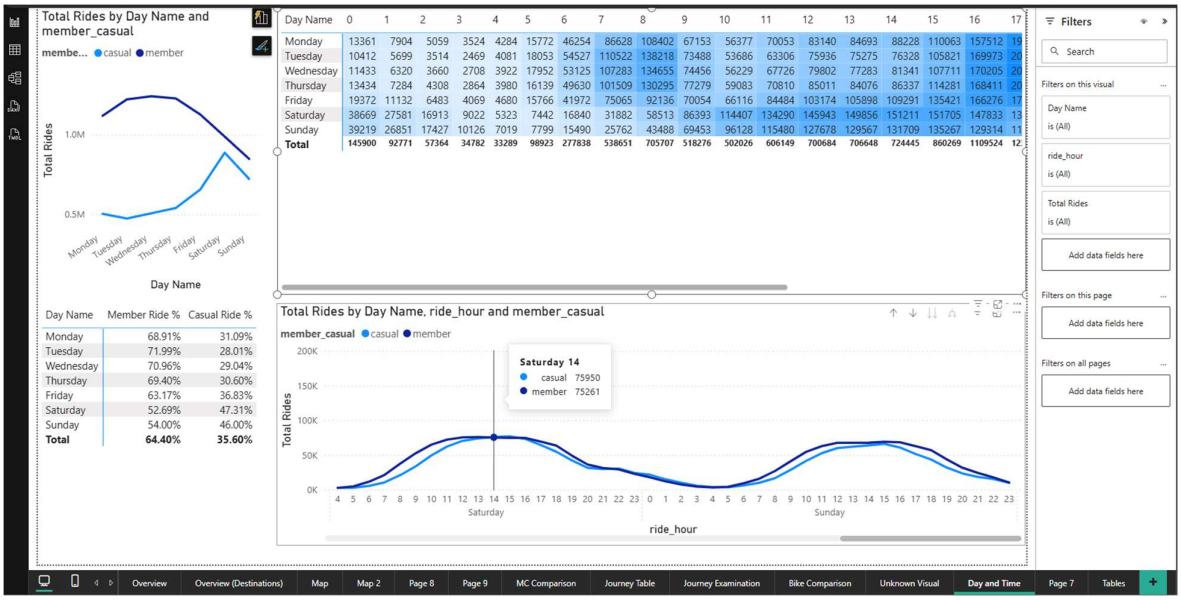
The percentage heatmap should be interpreted as the composition of each day/hour cell, not as each rider group's share of all weekly rides. For example, the Saturday afternoon cells approach 50% casual, whereas weekday morning cells are much more strongly member-dominated.



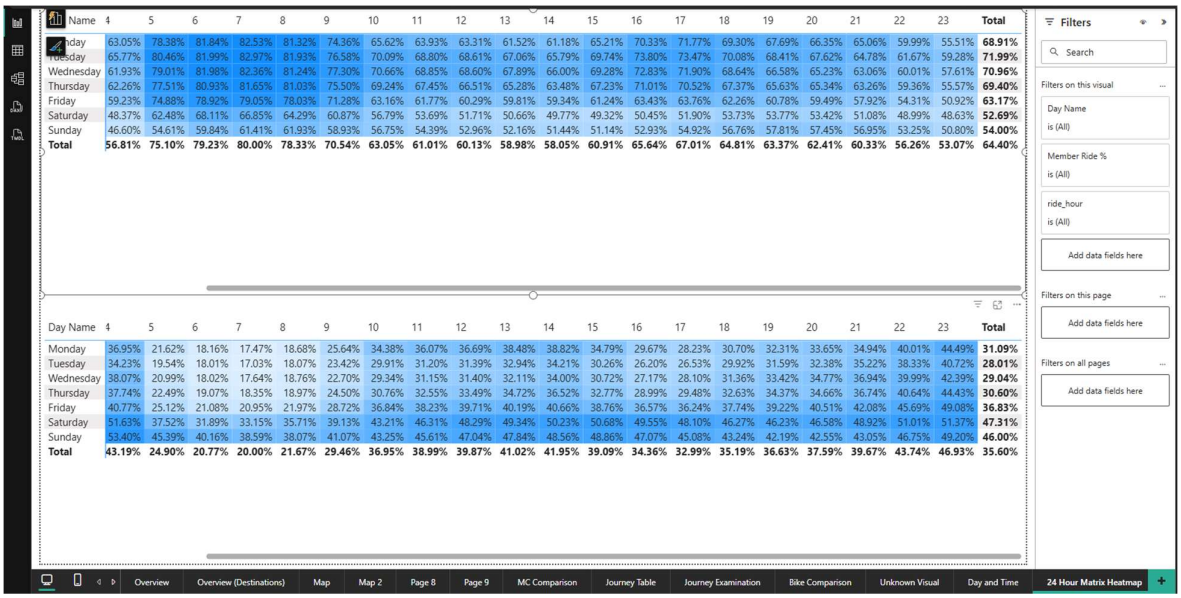
Day and Time.png



Day and Time 2.png



Day and Time 3.png



24 Hour Matrix Heatmap.png

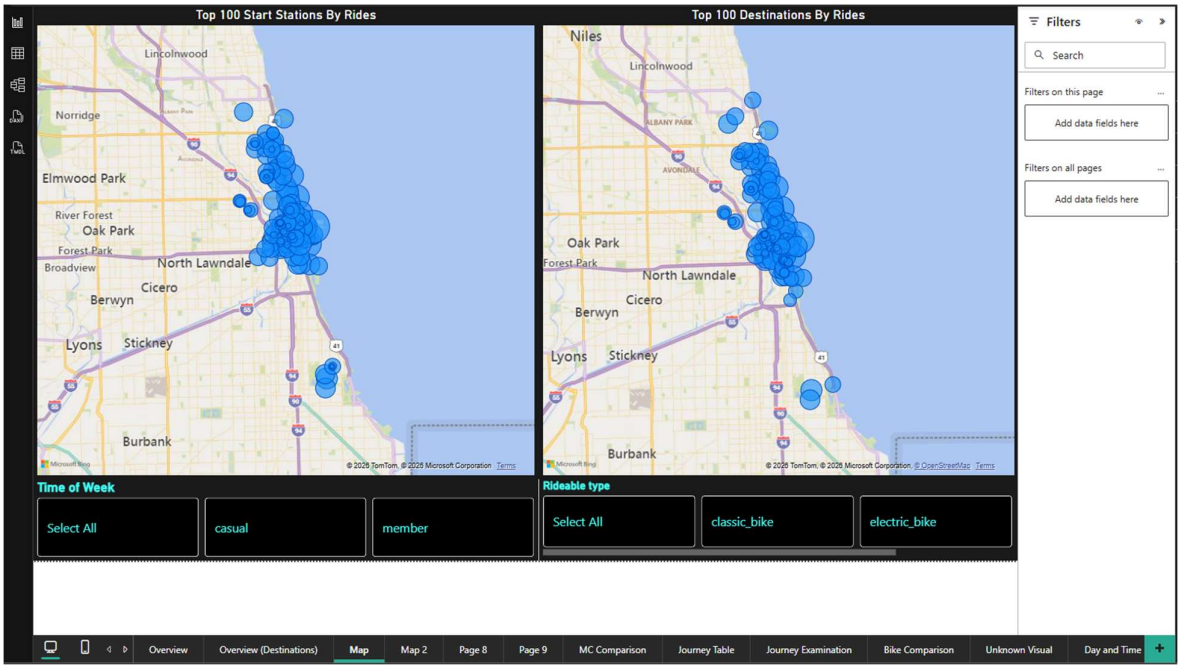


24 Hour Matrix Heatmap 2(1).png

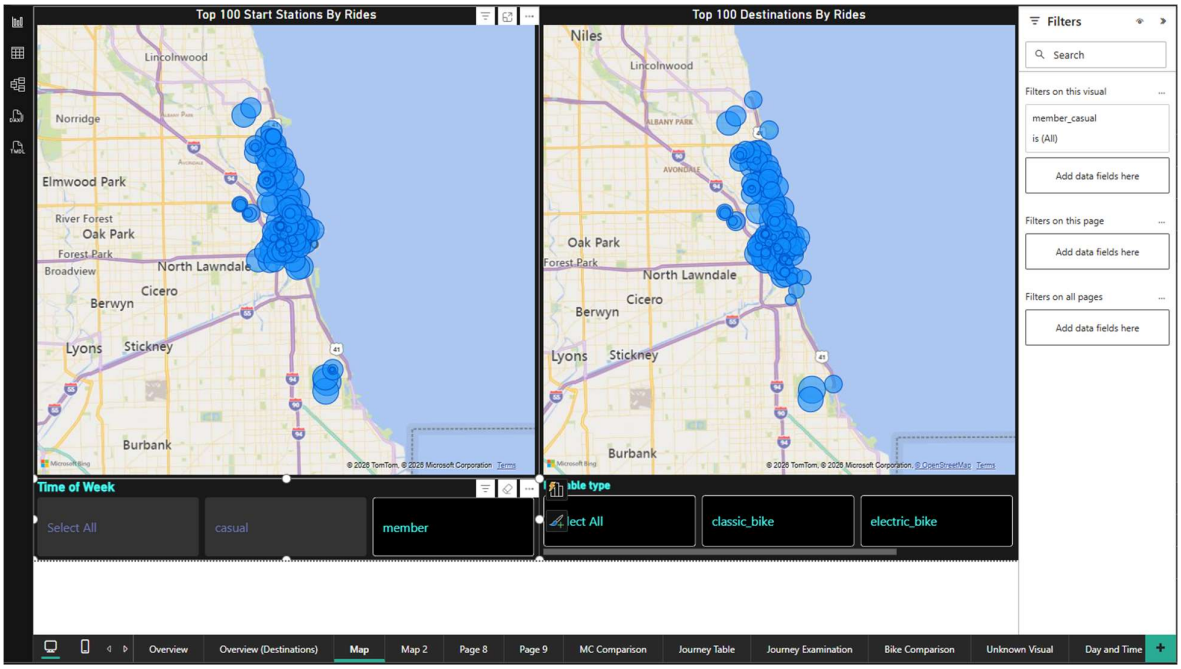
4. Station-Level Casual Concentration

The station analysis shows that casual rides are concentrated at a different set of locations from the strongest member stations. High casual-volume stations include Streeter Dr & Grand Ave, DuSable Lake Shore Dr & Monroe St, Michigan Ave & Oak St, DuSable Lake Shore Dr & North Blvd, Millennium Park, Shedd Aquarium, DuSable Harbor, Theater on the Lake, and Navy Pier.

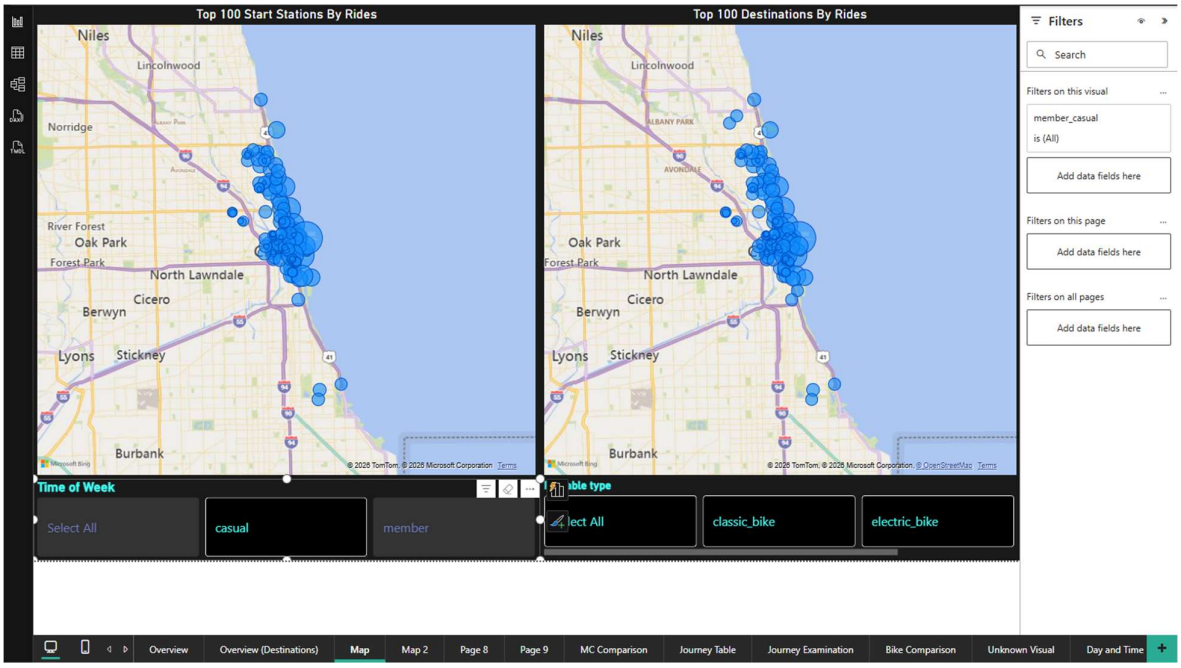
Many of these locations are associated with major attractions, the lakefront, or recreational areas. This is an observed geographic pattern; the station data alone cannot establish the purpose of an individual ride or prove that tourism caused the higher casual share.



Map.png



Map (Members).png



Map (Casuals).png

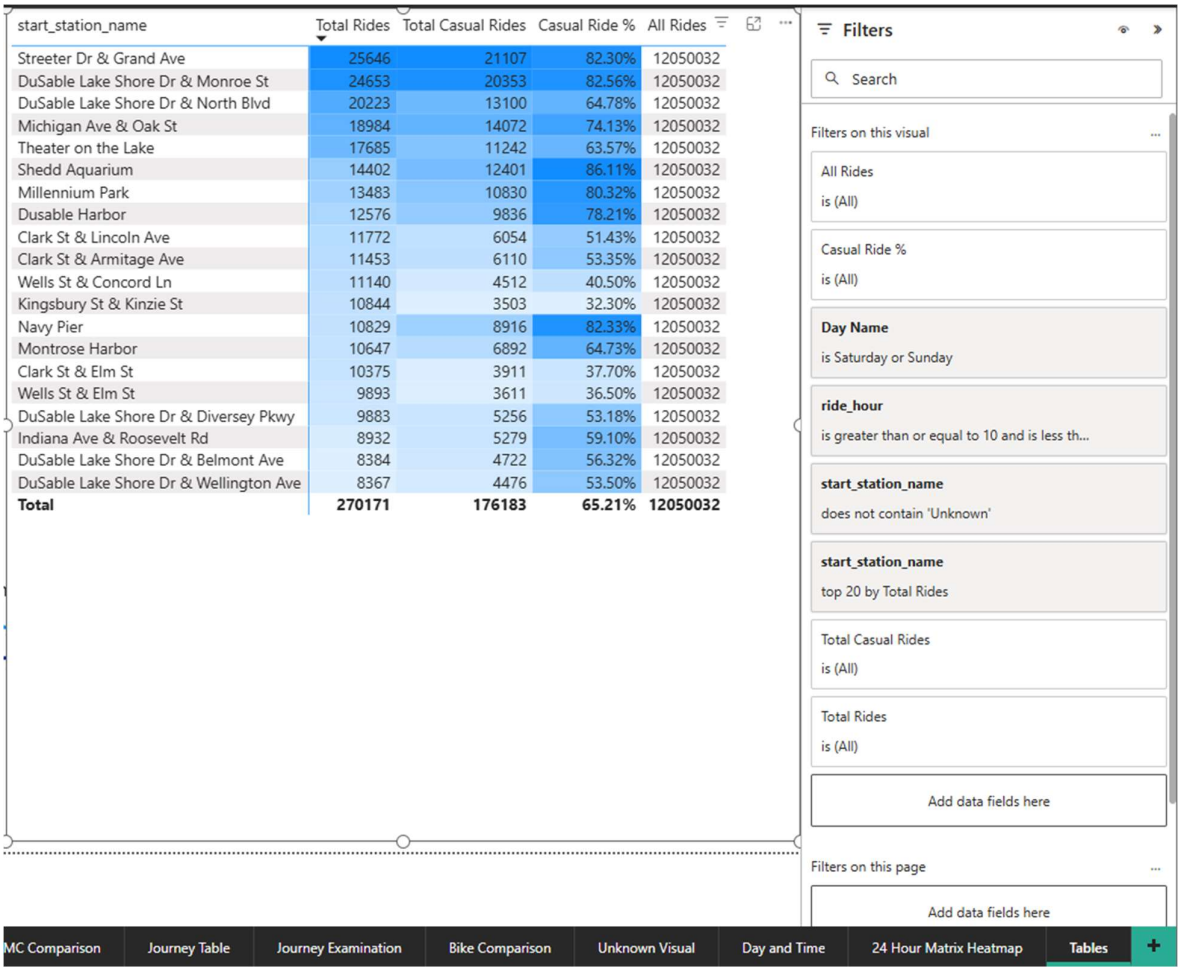
5. Warm-Month Weekend Daytime Station Analysis

A final cross-analysis was used to test whether stations with high casual participation become particularly casual-heavy during warmer-month weekend daytime periods.

Under the selected filters, the analysis contained 270,171 rides, of which 176,183 were casual, giving a 65.21% casual share. This is substantially above the approximately 35.4% casual share across the broader analysis period.

Several high-volume stations were particularly casual-heavy in this filtered period. Examples include Shedd Aquarium (86.11% casual), DuSable Lake Shore Dr & Monroe St (82.56%), Navy Pier (82.33%), Streeter Dr & Grand Ave (82.30%), Millennium Park (80.32%), and DuSable Harbor (78.21%).

This pattern identifies a potential opportunity for membership-conversion activity during warmer-month weekend daytime periods, particularly around high-volume stations with high casual shares. It should be treated as an opportunity suggested by the data rather than proof that such a campaign would cause conversions.



start_station_name	Total Rides	Total Casual Rides	Casual Ride %	All Rides
Streeter Dr & Grand Ave	25646	21107	82.30%	12050032
DuSable Lake Shore Dr & Monroe St	24653	20353	82.56%	12050032
DuSable Lake Shore Dr & North Blvd	20223	13100	64.78%	12050032
Michigan Ave & Oak St	18984	14072	74.13%	12050032
Theater on the Lake	17685	11242	63.57%	12050032
Shedd Aquarium	14402	12401	86.11%	12050032
Millennium Park	13483	10830	80.32%	12050032
Dusable Harbor	12576	9836	78.21%	12050032
Clark St & Lincoln Ave	11772	6054	51.43%	12050032
Clark St & Armitage Ave	11453	6110	53.35%	12050032
Wells St & Concord Ln	11140	4512	40.50%	12050032
Kingsbury St & Kinzie St	10844	3503	32.30%	12050032
Navy Pier	10829	8916	82.33%	12050032
Montrose Harbor	10647	6892	64.73%	12050032
Clark St & Elm St	10375	3911	37.70%	12050032
Wells St & Elm St	9893	3611	36.50%	12050032
DuSable Lake Shore Dr & Diversey Pkwy	9883	5256	53.18%	12050032
Indiana Ave & Roosevelt Rd	8932	5279	59.10%	12050032
DuSable Lake Shore Dr & Belmont Ave	8384	4722	56.32%	12050032
DuSable Lake Shore Dr & Wellington Ave	8367	4476	53.50%	12050032
Total	**270171**	**176183**	**65.21%**	**12050032**

Key insight table (Casual).png

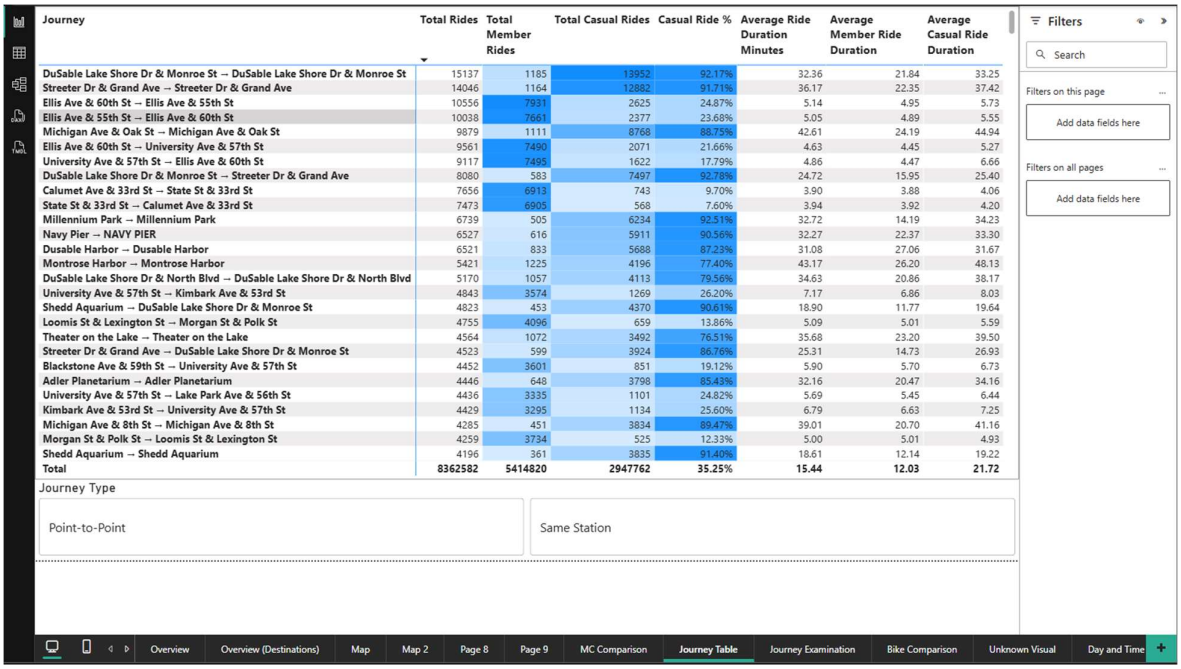
6. Journey Type Analysis

Journey behaviour was examined by comparing point-to-point trips with trips that started and ended at the same station.

Among the rides classified into the identified journey categories, approximately 94.8% were point-to-point and 5.2% were same-station. Point-to-point rides were approximately 66.24% member and 33.76% casual. Same-station rides were approximately 62.23% casual and 37.77% member.

The most common individual journeys included several same-station journeys at major lakefront and attraction locations. For example, DuSable Lake Shore Dr & Monroe St → the same station accounted for 15,137 rides, with a 92.17% casual share, while Streeter Dr & Grand Ave → the same station accounted for 14,046 rides, with a 91.71% casual share.

Same-station journeys may represent rides in which users return to their starting point, but the data does not record rider intent. Therefore, this is best treated as an interesting behavioural pattern rather than a confirmed explanation. Further analysis by hour and day did not reveal a strong additional pattern that justified making journey type a major business recommendation.



Journey	Total Rides	Total Member Rides	Total Casual Rides	Casual Ride %	Average Ride Duration Minutes	Average Member Ride Duration	Average Casual Ride Duration
DuSable Lake Shore Dr & Monroe St → DuSable Lake Shore Dr & Monroe St	15137	1185	13952	92.17%	32.36	21.84	33.25
Streeter Dr & Grand Ave → Streeter Dr & Grand Ave	14046	1164	12882	91.71%	36.17	22.35	37.42
Ellis Ave & 60th St → Ellis Ave & 55th St	10556	7931	2625	24.87%	5.14	4.95	5.73
Ellis Ave & 55th St → Ellis Ave & 60th St	10038	7661	2377	23.68%	5.05	4.89	5.55
Michigan Ave & Oak St → Michigan Ave & Oak St	9879	1111	8768	88.75%	42.61	24.19	44.94
Ellis Ave & 60th St → University Ave & 57th St	9561	7490	2071	21.66%	4.63	4.45	5.27
University Ave & 57th St → Ellis Ave & 60th St	9117	7495	1622	17.79%	4.86	4.47	6.66
DuSable Lake Shore Dr & Monroe St → Streeter Dr & Grand Ave	8080	583	7497	92.78%	24.72	15.95	25.40
Calumet Ave & 33rd St → State St & 33rd St	7656	6913	743	9.70%	3.90	3.88	4.06
State St & 33rd St → Calumet Ave & 33rd St	7473	6905	568	7.60%	3.94	3.92	4.20
Millennium Park → Millennium Park	6739	505	6234	92.51%	32.72	14.19	34.23
Navy Pier → NAVY PIER	6527	616	5911	90.56%	32.27	22.37	33.30
Dusable Harbor → Dusable Harbor	6521	833	5688	87.23%	31.08	27.06	31.67
Montrose Harbor → Montrose Harbor	5421	1225	4196	77.40%	43.17	26.20	48.13
DuSable Lake Shore Dr & North Blvd → DuSable Lake Shore Dr & North Blvd	5170	1057	4113	79.56%	34.63	20.86	38.17
University Ave & 57th St → Kimbark Ave & 53rd St	4843	3574	1269	26.20%	7.17	6.86	8.03
Shedd Aquarium → DuSable Lake Shore Dr & Monroe St	4823	453	4370	90.61%	18.90	11.77	19.64
Loomis St & Lexington St → Morgan St & Polk St	4755	4096	659	13.86%	5.09	5.01	5.59
Theater on the Lake → Theater on the Lake	4564	1072	3492	76.51%	35.68	23.20	39.50
Streeter Dr & Grand Ave → DuSable Lake Shore Dr & Monroe St	4523	599	3924	86.76%	25.31	14.73	26.93
Blackstone Ave & 59th St → University Ave & 57th St	4452	3601	851	19.12%	5.90	5.70	6.73
Adler Planetarium → Adler Planetarium	4446	648	3798	85.43%	32.16	20.47	34.16
University Ave & 57th St → Lake Park Ave & 56th St	4436	3335	1101	24.82%	5.69	5.45	6.44
Kimbark Ave & 53rd St → University Ave & 57th St	4429	3295	1134	25.60%	6.79	6.63	7.25
Michigan Ave & 8th St → Michigan Ave & 8th St	4285	451	3834	89.47%	39.01	20.70	41.16
Morgan St & Polk St → Loomis St & Lexington St	4259	3734	525	12.33%	5.00	5.01	4.93
Shedd Aquarium → Shedd Aquarium	4196	361	3835	91.40%	18.61	12.14	19.22
Total	**8362582**	**5414820**	**2947762**	**35.25%**	**15.44**	**12.03**	**21.72**

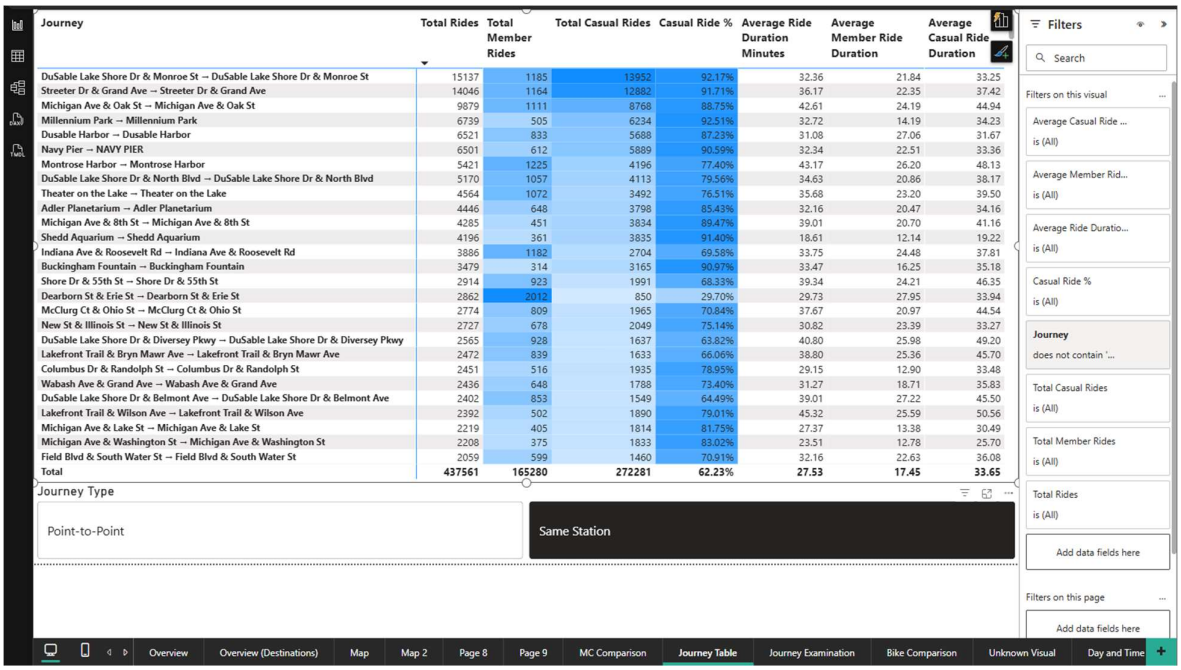
Journey Type

Point-to-Point | Same Station

Filters | Search | Filters on this page | Add data fields here | Filters on all pages | Add data fields here

Overview | Overview (Destinations) | Map | Map 2 | Page 8 | Page 9 | MC Comparison | Journey Table | Journey Examination | Bike Comparison | Unknown Visual | Day and Time

Journey Table.png



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Millennium Park → Millennium Park	6739	505	6234	92.51%	32.72	14.19	34.23
Dusable Harbor → Dusable Harbor	6521	833	5688	87.23%	31.08	27.06	31.67
Navy Pier → NAVY PIER	6501	612	5889	90.59%	32.34	22.51	33.36
Montrose Harbor → Montrose Harbor	5421	1225	4196	77.40%	43.17	26.20	48.13
DuSable Lake Shore Dr & North Blvd → DuSable Lake Shore Dr & North Blvd	5170	1057	4113	79.56%	34.63	20.86	38.17
Theater on the Lake → Theater on the Lake	4564	1072	3492	76.51%	35.68	23.20	39.50
Adler Planetarium → Adler Planetarium	4446	648	3798	85.43%	32.16	20.47	34.16
Michigan Ave & 8th St → Michigan Ave & 8th St	4285	451	3834	89.47%	39.01	20.70	41.16
Shedd Aquarium → Shedd Aquarium	4196	361	3835	91.40%	18.61	12.14	19.22
Indiana Ave & Roosevelt Rd → Indiana Ave & Roosevelt Rd	3886	1182	2704	69.58%	33.75	24.48	37.81
Buckingham Fountain → Buckingham Fountain	3479	314	3165	90.97%	33.47	16.25	35.18
Shore Dr & 55th St → Shore Dr & 55th St	2914	923	1991	68.33%	39.34	24.21	46.35
Dearborn St & Erie St → Dearborn St & Erie St	2862	2012	850	29.70%	29.73	27.95	33.94
McClurg Ct & Ohio St → McClurg Ct & Ohio St	2774	809	1965	70.84%	37.67	20.97	44.54
New St & Illinois St → New St & Illinois St	2727	678	2049	75.14%	30.82	23.39	33.27
DuSable Lake Shore Dr & Diversey Pkwy → DuSable Lake Shore Dr & Diversey Pkwy	2565	928	1637	63.82%	40.80	25.98	49.20
Lakefront Trail & Bryn Mawr Ave → Lakefront Trail & Bryn Mawr Ave	2472	839	1633	66.06%	38.80	25.36	45.70
Columbus Dr & Randolph St → Columbus Dr & Randolph St	2451	516	1935	78.95%	29.15	12.90	33.48
Wabash Ave & Grand Ave → Wabash Ave & Grand Ave	2436	648	1788	73.40%	31.27	18.71	35.83
DuSable Lake Shore Dr & Belmont Ave → DuSable Lake Shore Dr & Belmont Ave	2402	853	1549	64.49%	39.01	27.22	45.50
Lakefront Trail & Wilson Ave → Lakefront Trail & Wilson Ave	2392	502	1890	79.01%	45.32	25.59	50.56
Michigan Ave & Lake St → Michigan Ave & Lake St	2219	405	1814	81.75%	27.37	13.38	30.49
Michigan Ave & Washington St → Michigan Ave & Washington St	2208	375	1833	83.02%	23.51	12.78	25.70
Field Blvd & South Water St → Field Blvd & South Water St	2059	599	1460	70.91%	32.16	22.63	36.08
Total	**437561**	**165280**	**272281**	**62.23%**	**27.53**	**17.45**	**33.65**

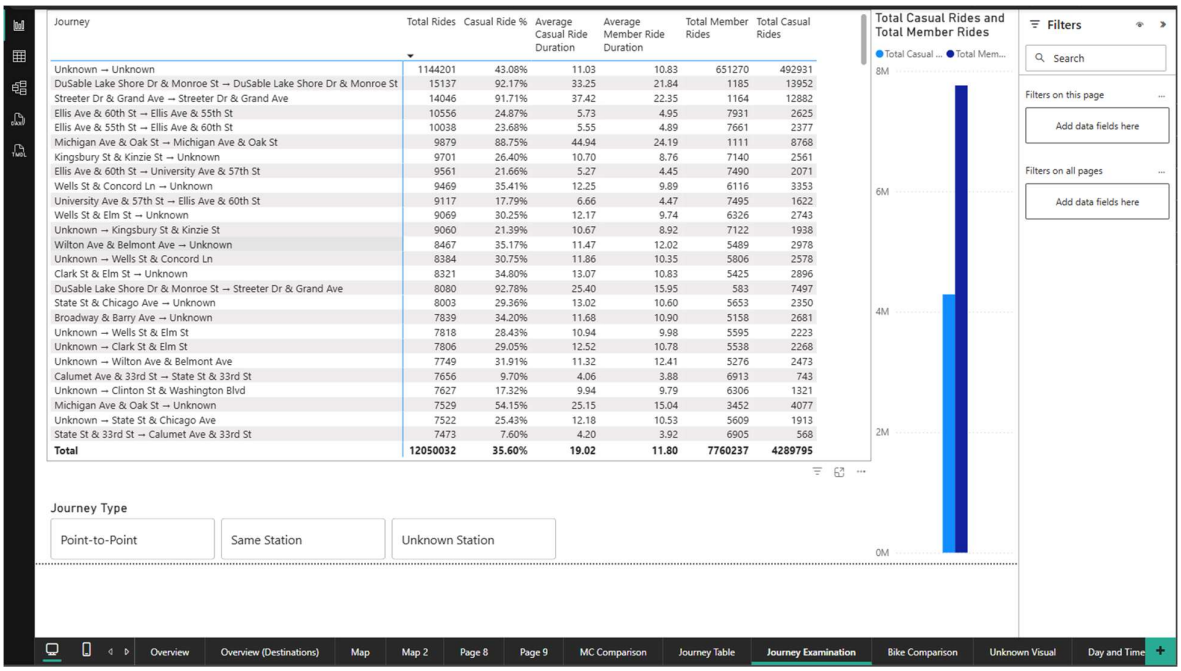
Journey Type

Point-to-Point | Same Station

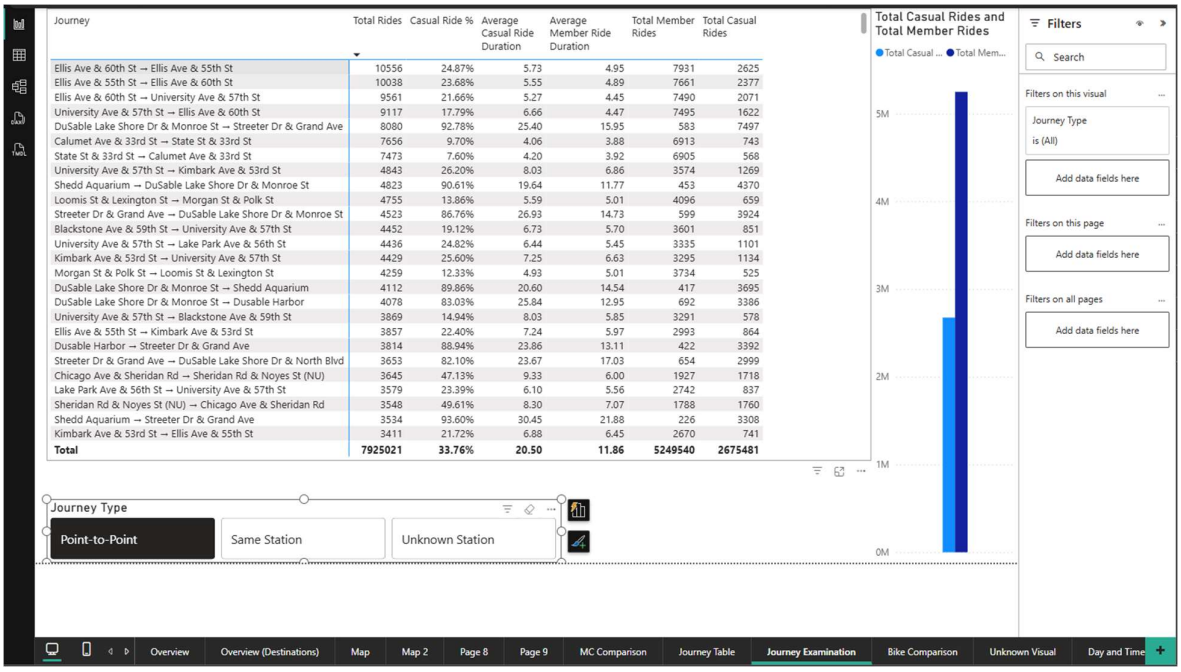
Filters | Search | Filters on this visual | Average Casual Ride ... is (All) | Average Member Rid... is (All) | Average Ride Duratio... is (All) | Casual Ride % is (All) | Journey does not contain '... | Total Casual Rides is (All) | Total Member Rides is (All) | Total Rides is (All) | Add data fields here | Filters on this page | Add data fields here

Overview | Overview (Destinations) | Map | Map 2 | Page 8 | Page 9 | MC Comparison | Journey Table | Journey Examination | Bike Comparison | Unknown Visual | Day and Time

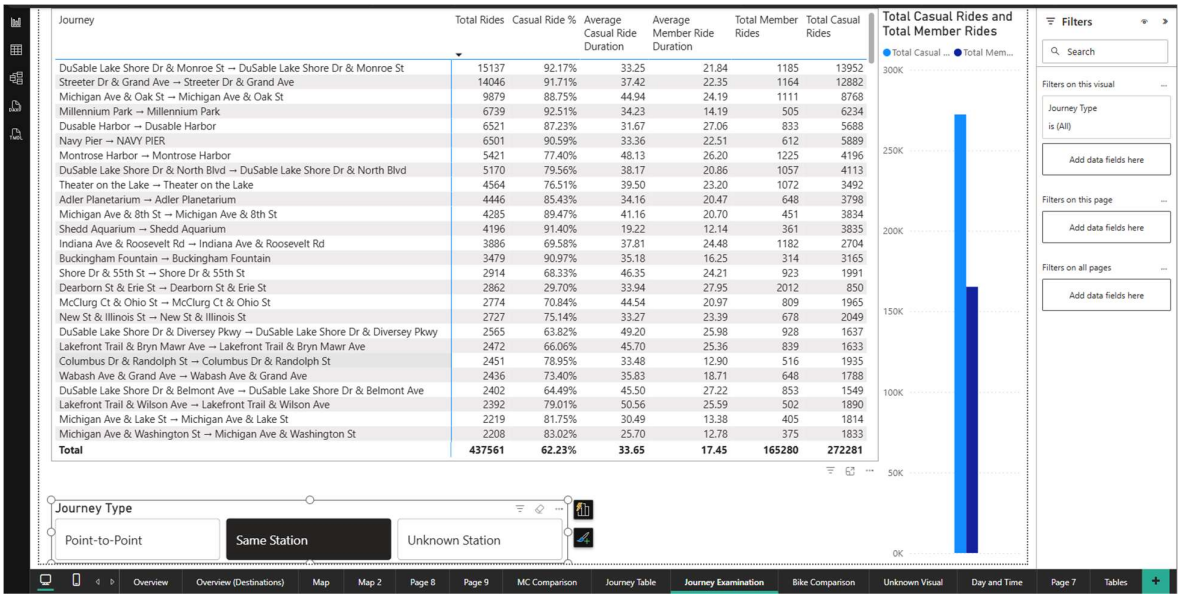
Journey Table Same Station.png



Journey Examination.png



Journey Examination Point-to-Point.png



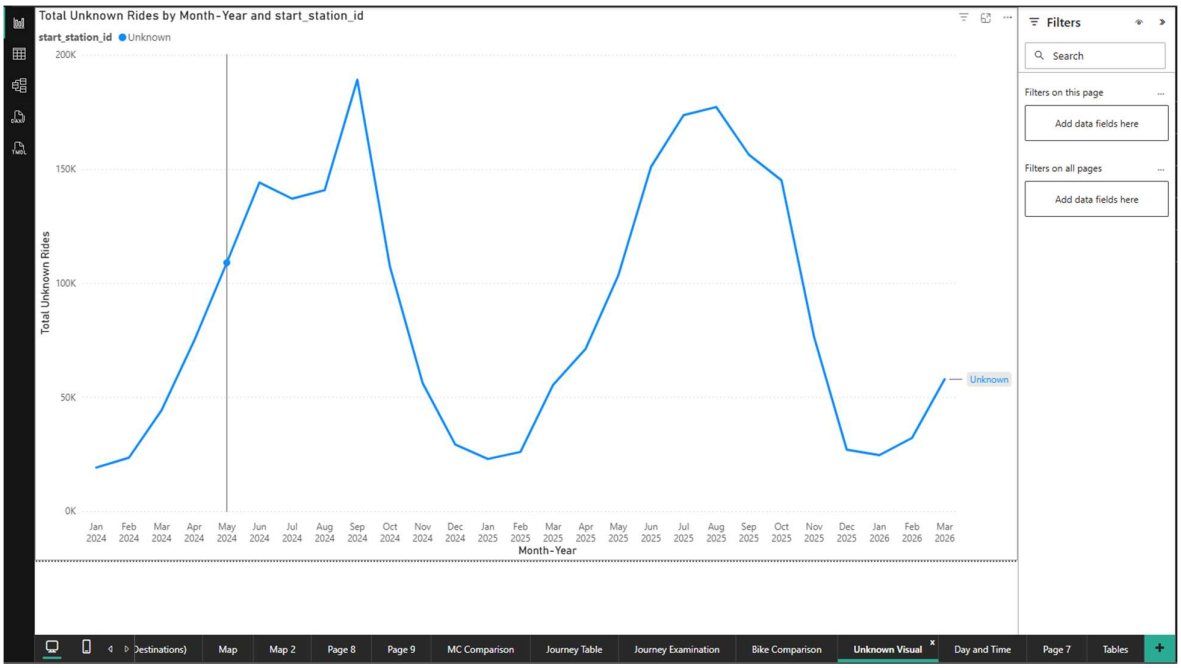
Journey Examination Same Station.png

7. Unknown Station Data Quality

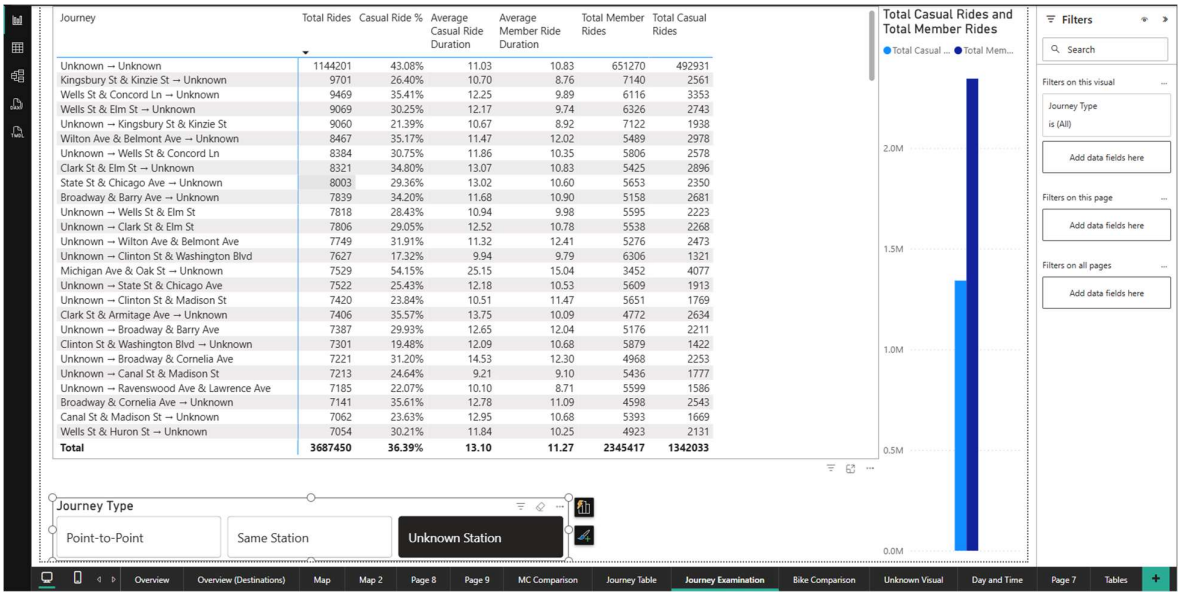
A substantial number of rides have no recorded station identity and were represented as 'Unknown'. Approximately 2.3 million rides in the broader analysis had an Unknown station, representing roughly 19% of the rides.

The Unknown-rides investigation examined whether these rides were concentrated at particular times. The monthly pattern rises and falls broadly with overall ride volume, rather than showing a distinctive time-specific anomaly. No strong evidence was found that Unknown rides were associated with a particular period of the day or a separate usage pattern.

Unknown rides were therefore retained in overall ride-level calculations, because removing them would discard a substantial portion of the dataset. For station-level analysis where a station identity is required, Unknown records were excluded.



Unknown Visual.png



Journey Examination Unknown Stations.png

8. Final Analysis Summary

The completed analysis shows several consistent differences between members and casual riders:

- Members generate more rides overall and show stronger weekday and commuting-hour activity.
- Members generally take shorter and more consistent rides.
- Casual riders generally take longer rides, with the difference becoming more pronounced during warmer months.

- Casual participation increases substantially during weekend daytime periods.
- Casual rides are concentrated around several central, lakefront, recreational, and attraction-associated stations.
- Electric-bike usage increased substantially during 2025 and into early 2026, with growth visible among both members and casual riders.
- Same-station journeys are a small proportion of total rides but have a notably higher casual share than point-to-point journeys.
- Unknown station records are a significant data-quality limitation for station-level analysis, but their timing does not show a distinctive pattern.

9. Potential Business Opportunity

The strongest combined pattern identified by the analysis is the concentration of casual riders during warmer-month weekend daytime periods, particularly around high-volume stations that also have high casual shares.

This suggests a potential opportunity to focus membership-conversion activity around those periods and locations. The analysis identifies where and when the casual audience is most concentrated, but it does not measure campaign response, membership conversion, pricing, revenue, or customer lifetime value. Those would be required to test the commercial impact of a specific campaign.

Electric-bike usage provides a separate potential area for testing membership benefits. The observed increase in electric-bike adoption and the slightly higher electric-bike share among casual riders make this worth investigating, but the current dataset does not establish which benefit structure would be commercially effective.

10. Limitations and Interpretation

- Station IDs were not stable across the analysis period, so station names were used to represent physical stations.
- Approximately 19% of rides had an Unknown station identity and therefore could not be assigned reliably to a named station.
- The analysis identifies associations and patterns; it does not establish causal relationships or rider intent.
- The dataset does not contain pricing, revenue, membership-conversion, campaign-response, or customer-lifetime-value measures needed to quantify the financial impact of proposed actions.
- Some journey patterns, such as same-station trips, can support hypotheses about rider behaviour but cannot by themselves establish why a ride occurred.

10. Status of Earlier Follow-Up Questions

The questions listed in the original notes were subsequently investigated. Electric-bike growth was broken down by rider type; station behaviour was compared by rider type; day-of-week and hour-of-day patterns were analysed; high-casual-share stations were examined; and the member-versus-casual difference was examined across seasons and years. The remaining journey and Unknown-station investigations did not produce additional evidence strong enough to warrant further analysis.